Appendix A - Abrolhos Shallow Bank Marine Park Natural Values Report

North-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
May 2021	2021-05 Abrolhos BOSS	BOSS	NPZ, SPZ	46	46	-	-
May 2021	2021-05 Abrolhos stereo-BRUVs	BRUV	NPZ, SPZ	24	24	23	23
Feb-Mar 2025	2025-02 Yamatji-Shallow-Bank stereo-BRUVs	BRUV	NPZ, SPZ	120	115	114	109

Benthic ecosystem extent

Observations

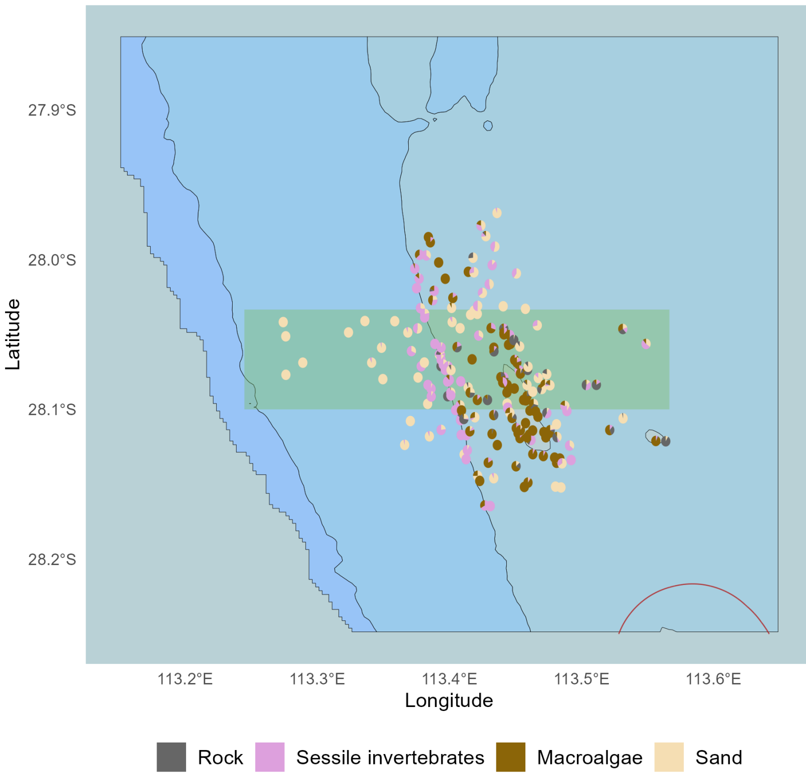

Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

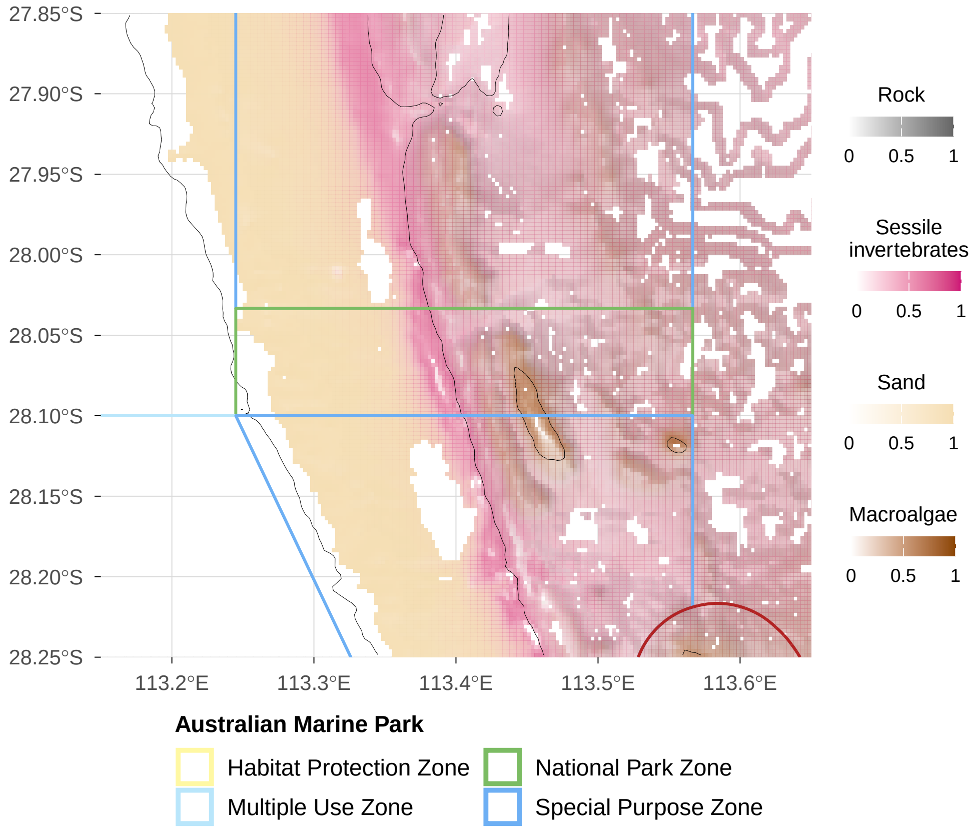

Figure 2: Predicted dominant habitat type from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

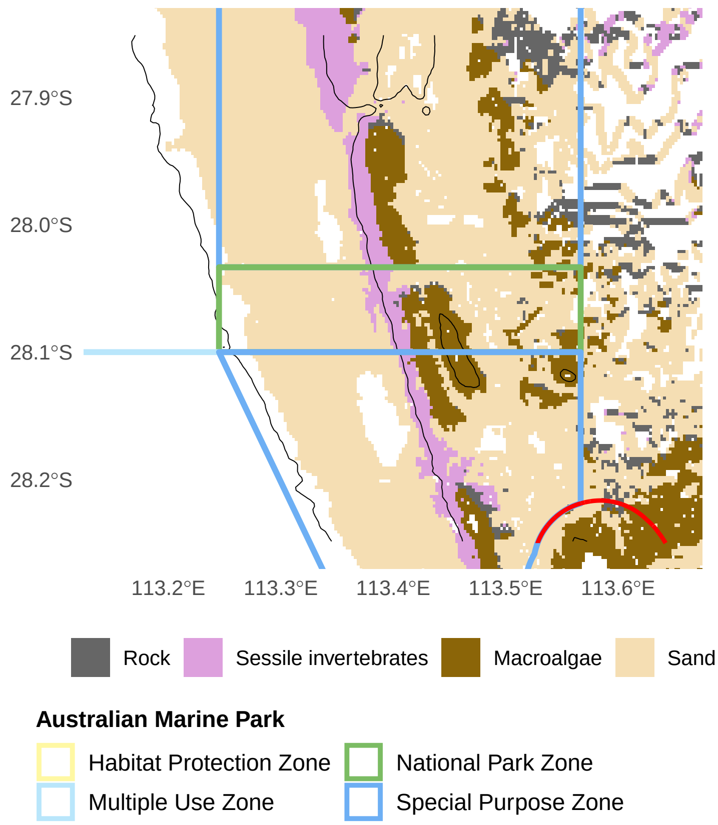

Figure 3: Predicted habitat class distribution from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

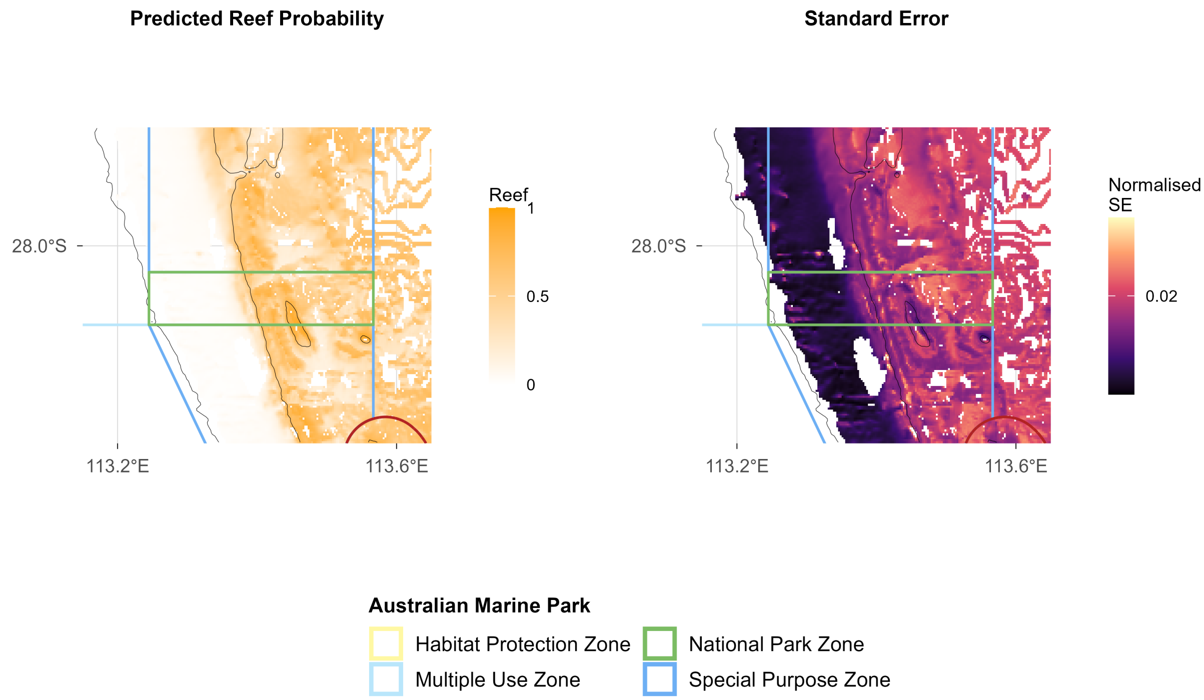

Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

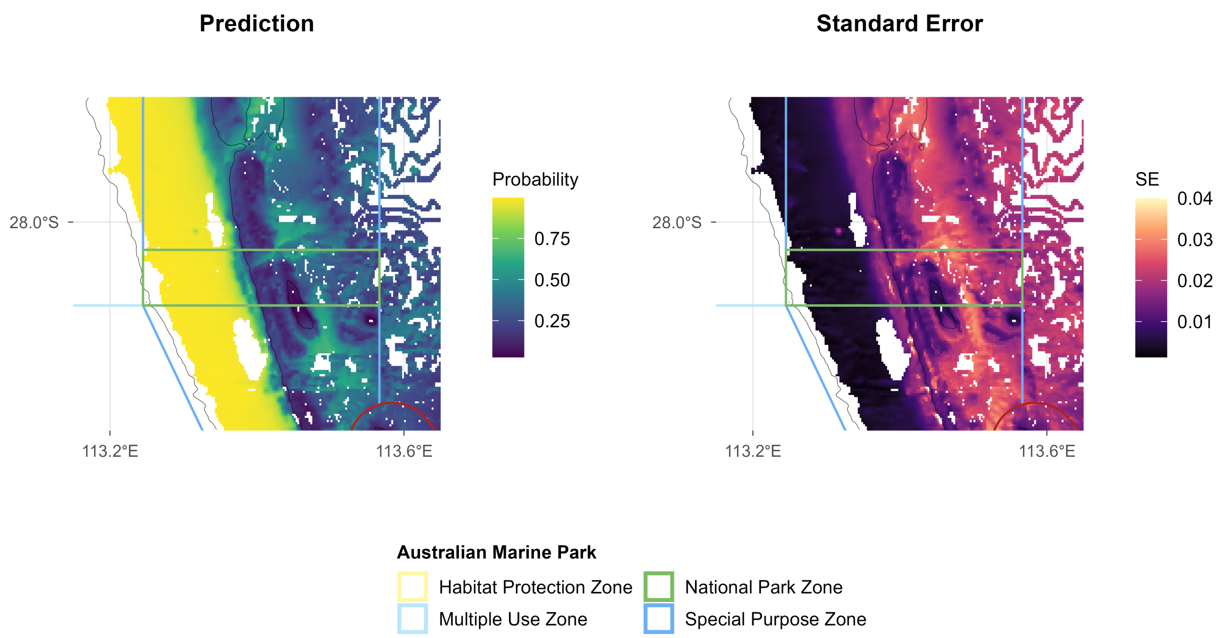

Figure 5: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

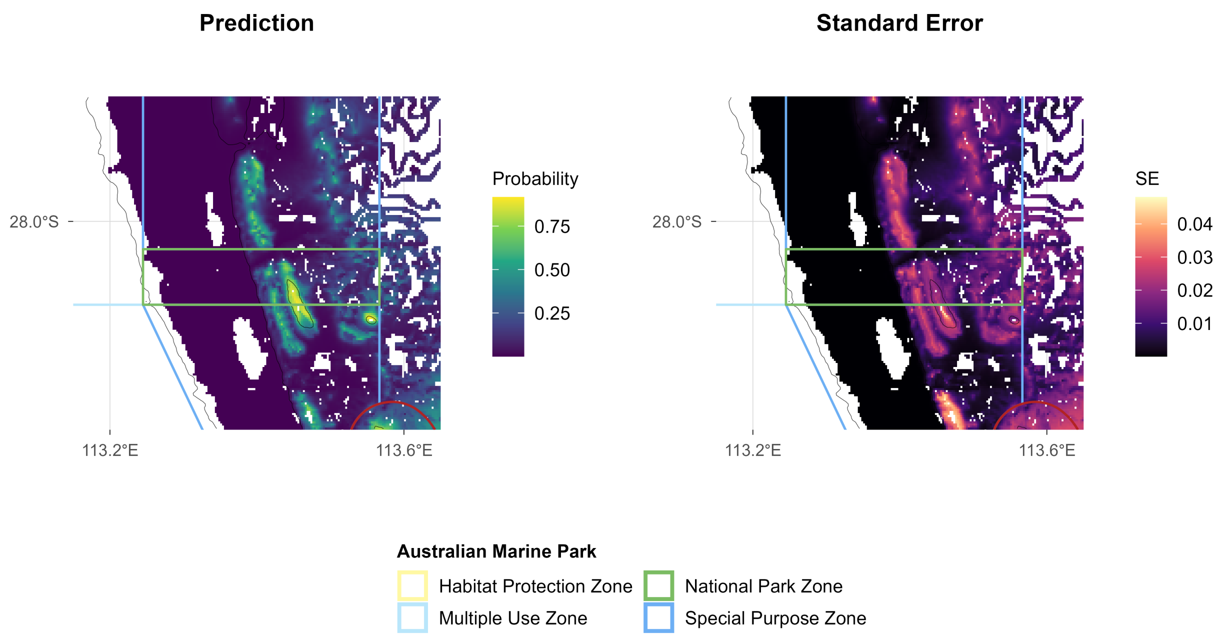

Figure 6: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Rock

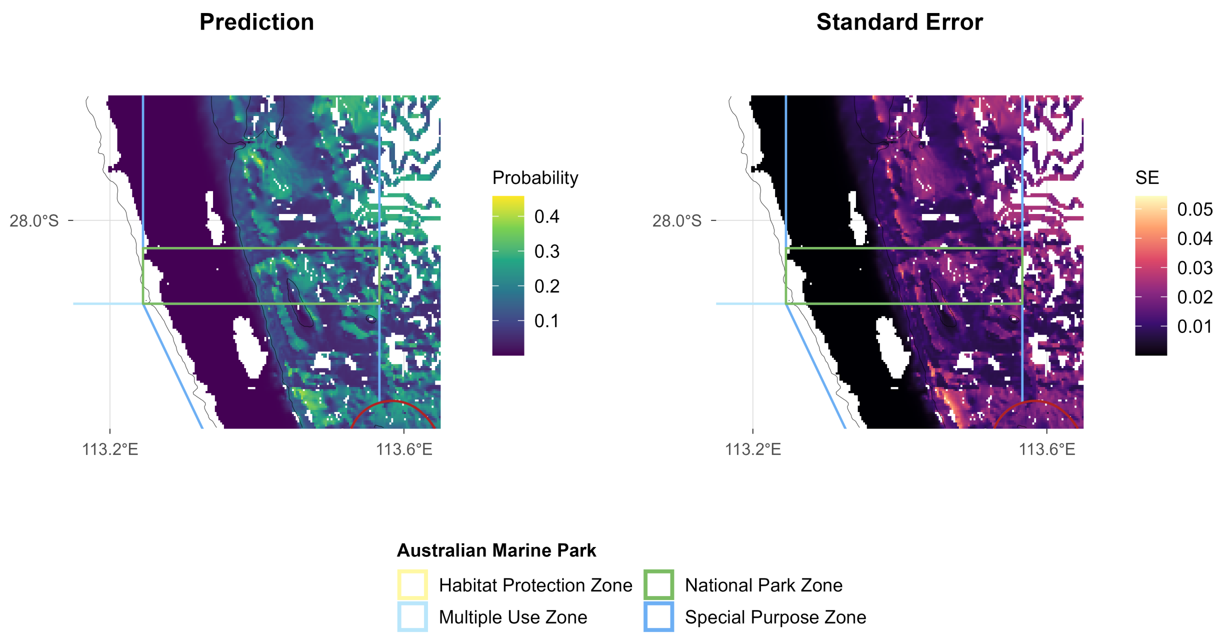

Figure 7: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

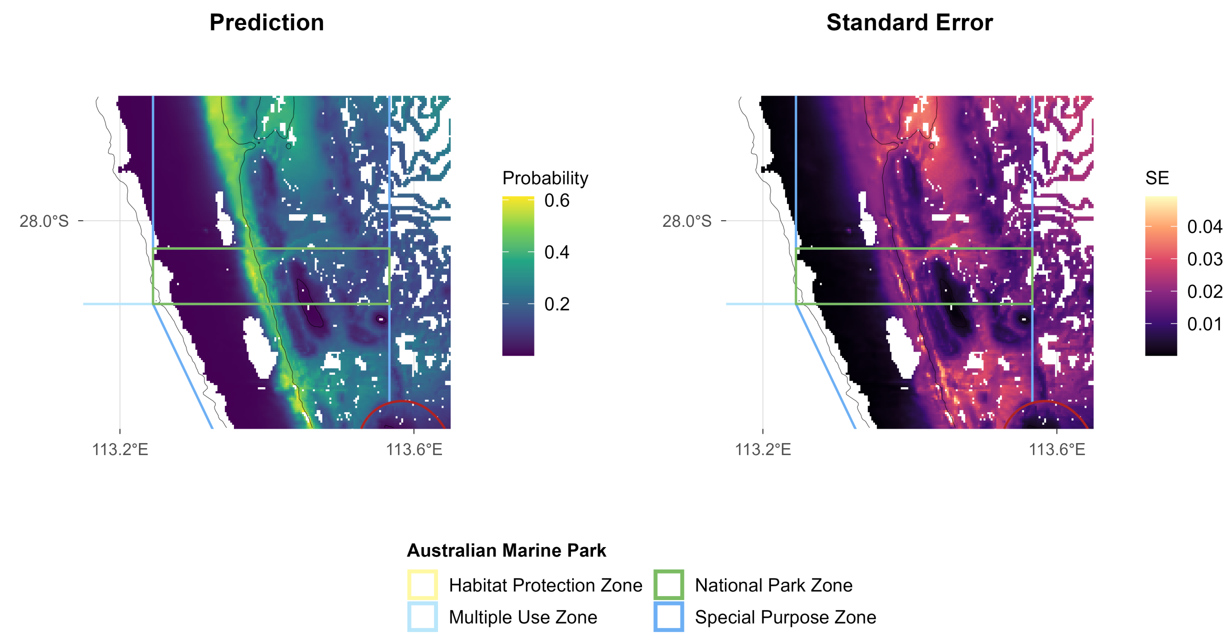

Figure 8: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across the 2021 and 2025 surveys. ABMP = Abrolhos Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

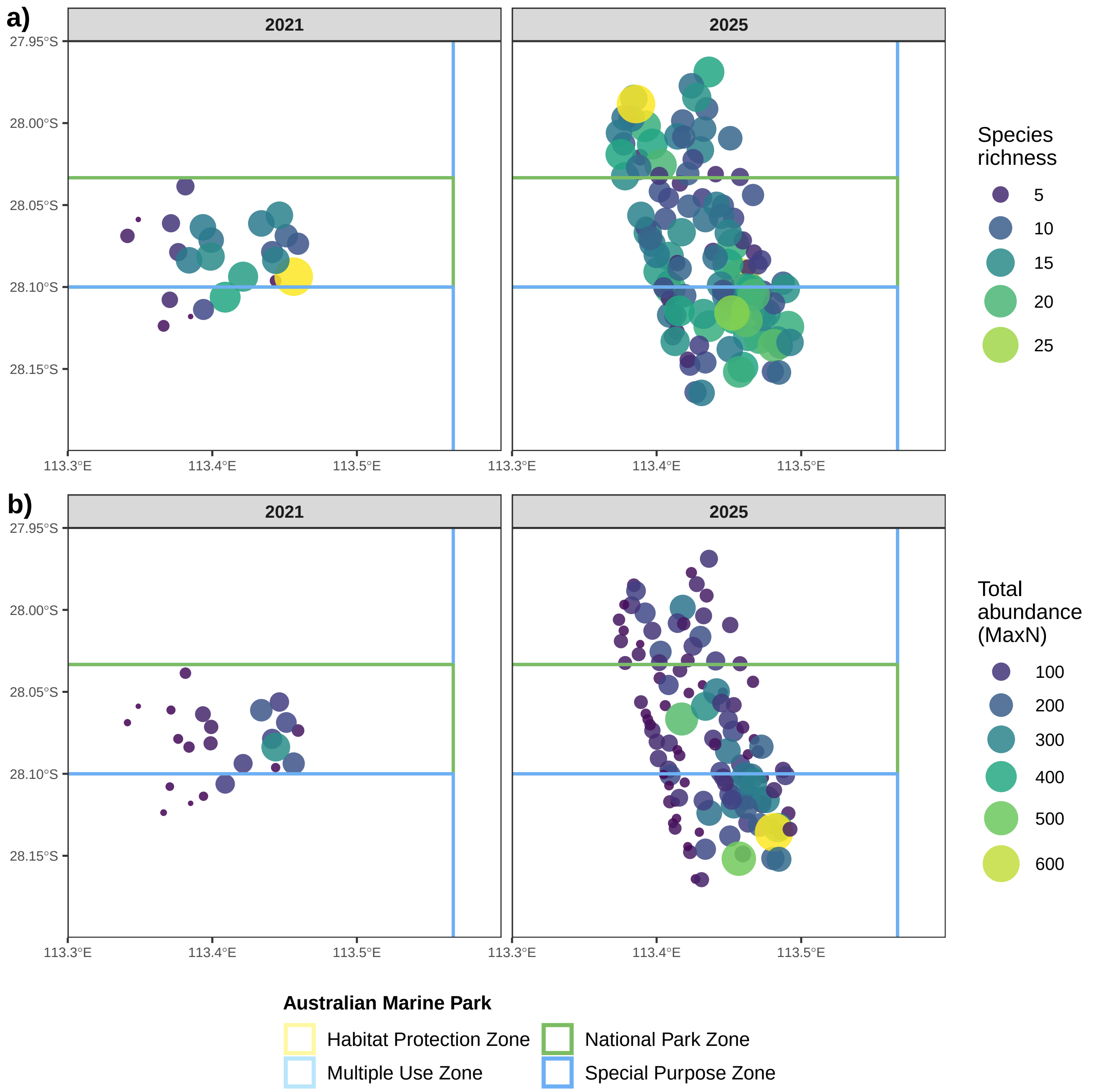

Figure 9: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV deployment, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

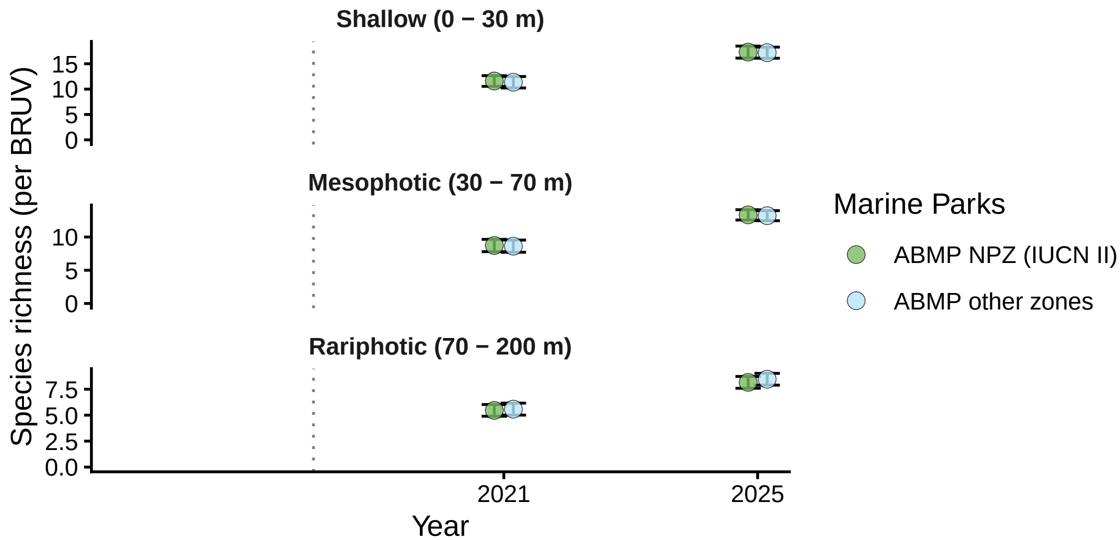

Figure 10.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

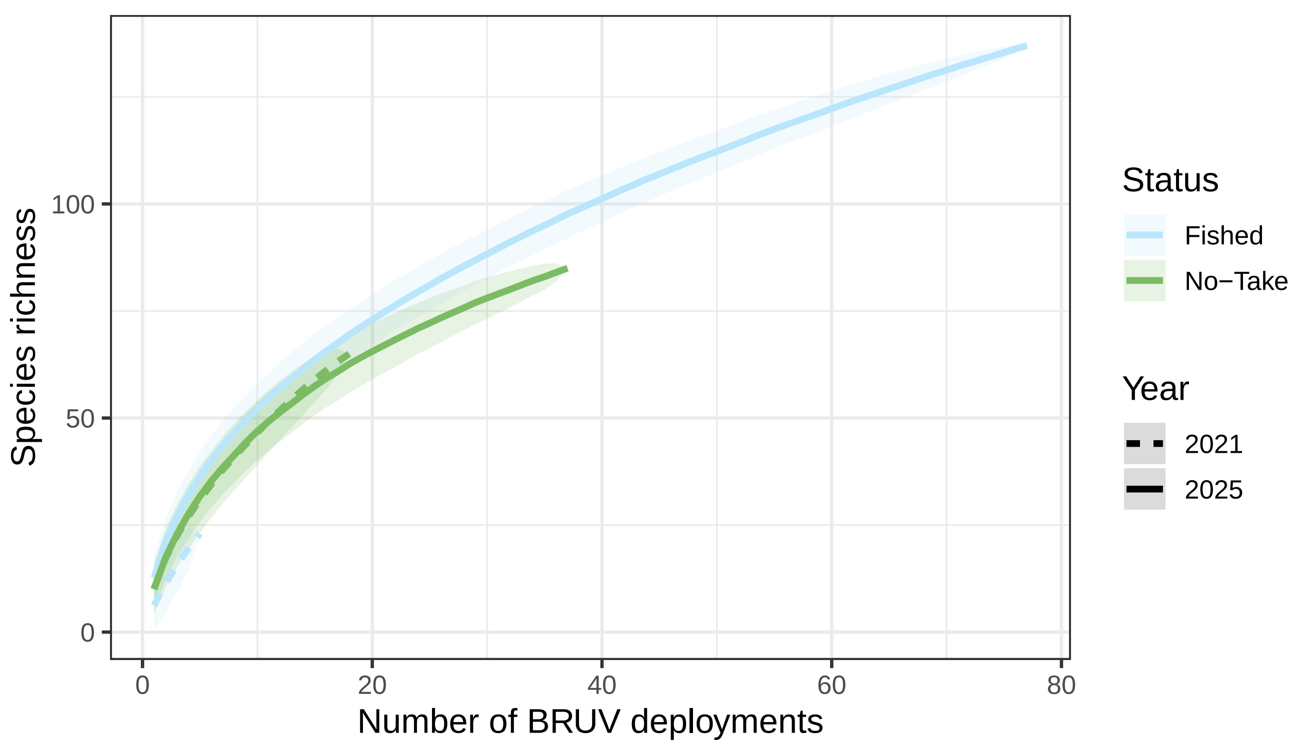

Figure 10.2: Demersal fish - Species richness: species accumulation curve by year and zoning status from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Species richness

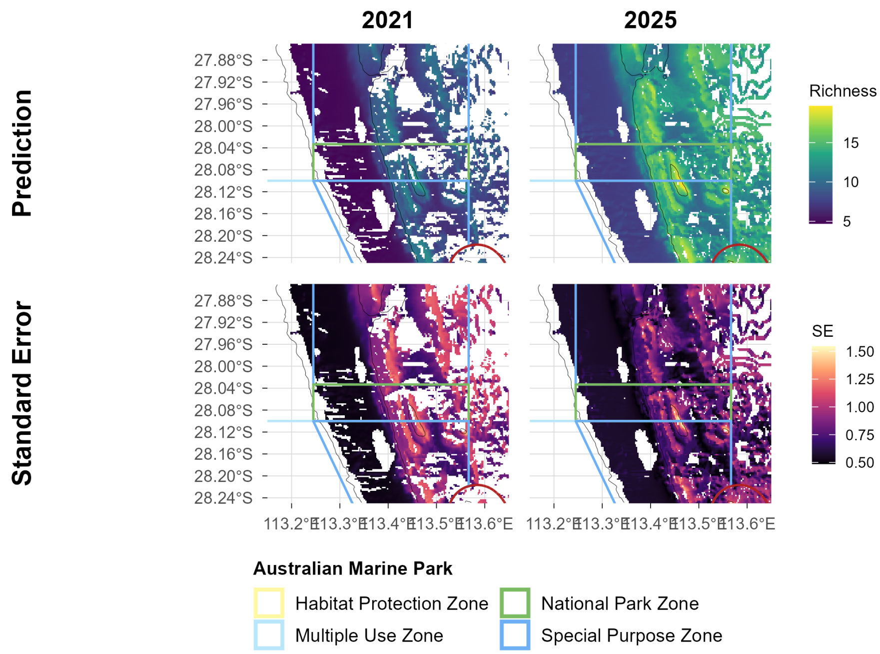

Figure 10.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

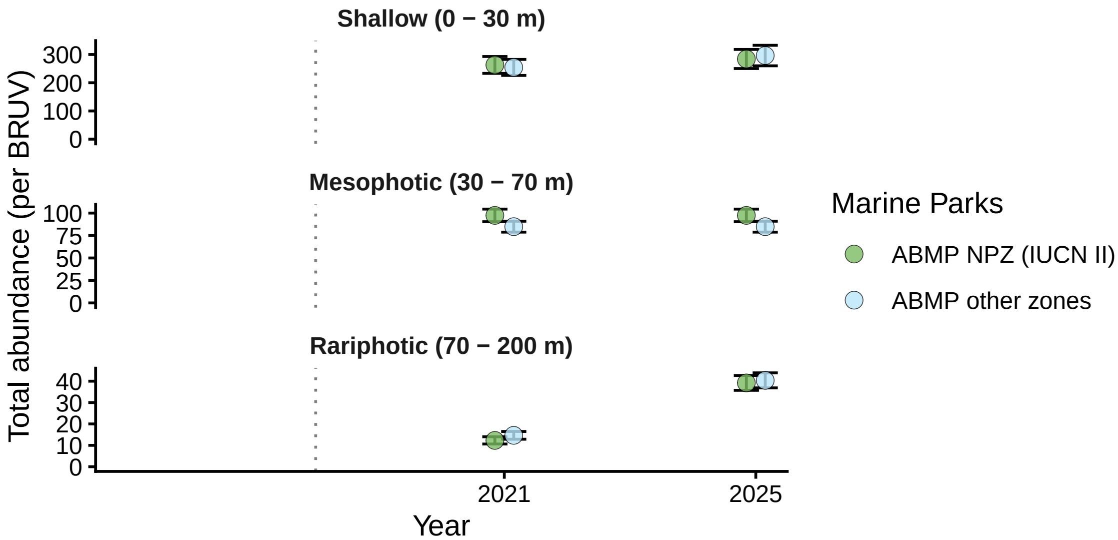

Figure 11.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

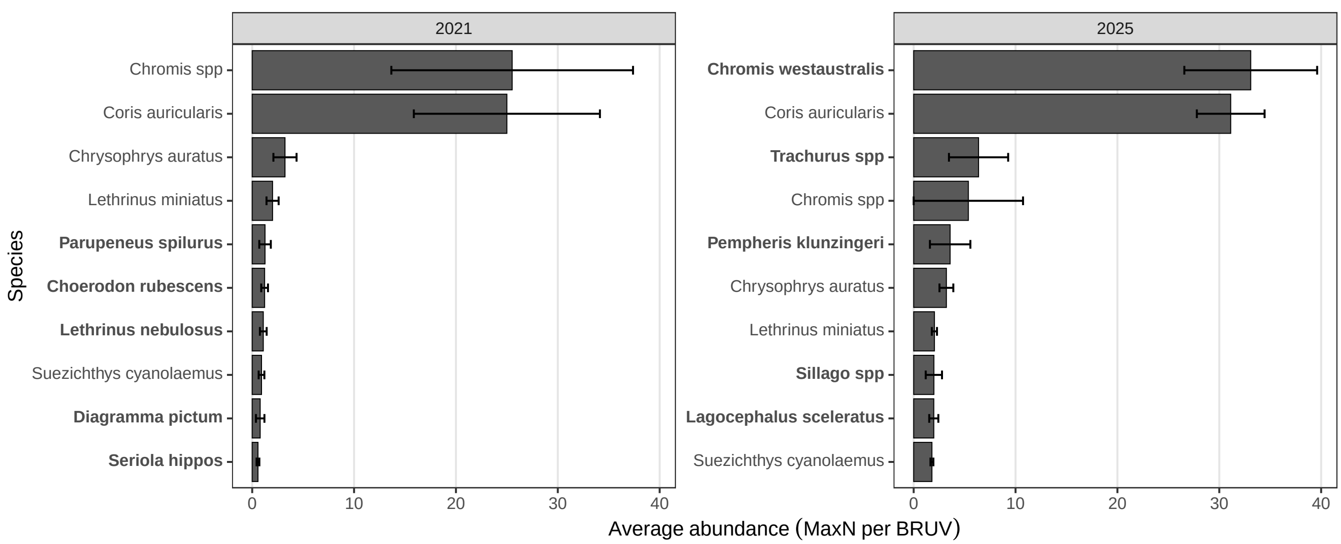

Figure 11.2: Demersal fish - Total abundance: top 10 most abundant species in each survey year from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Species appearing in only one year's top 10 are shown in bold.

Total abundance

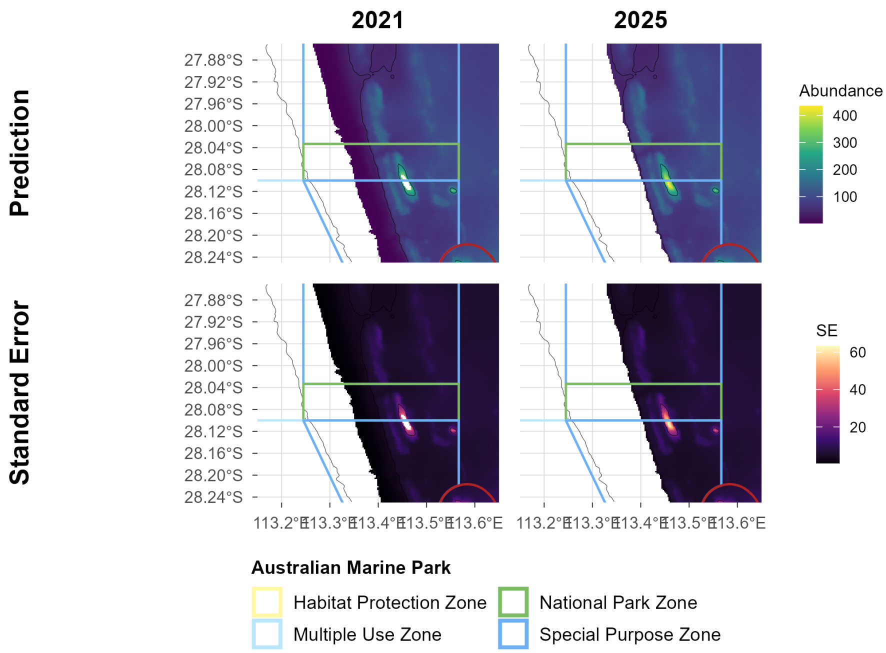

Figure 11.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

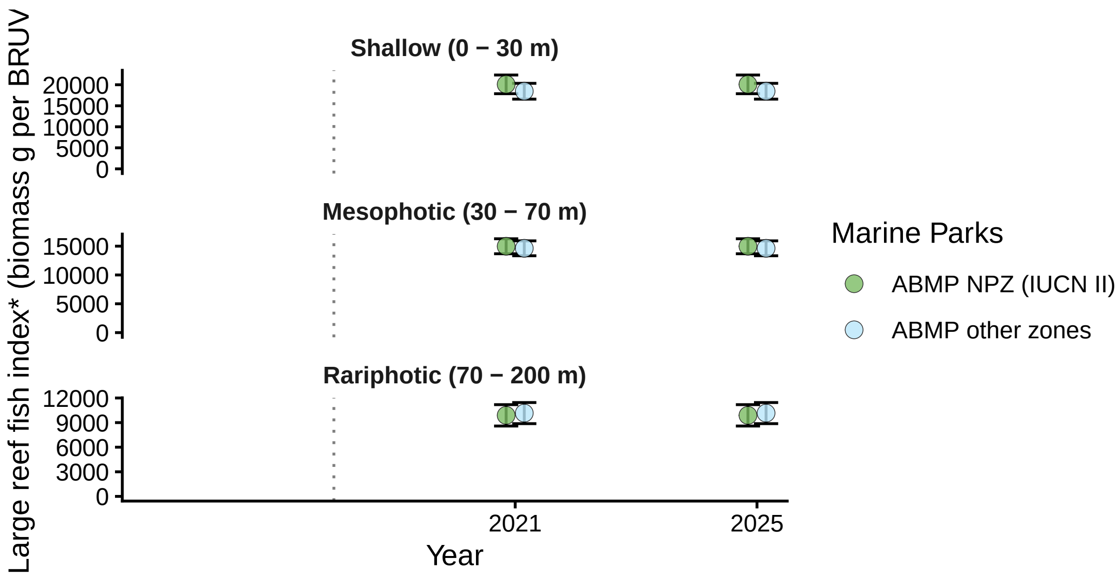

Figure 12.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

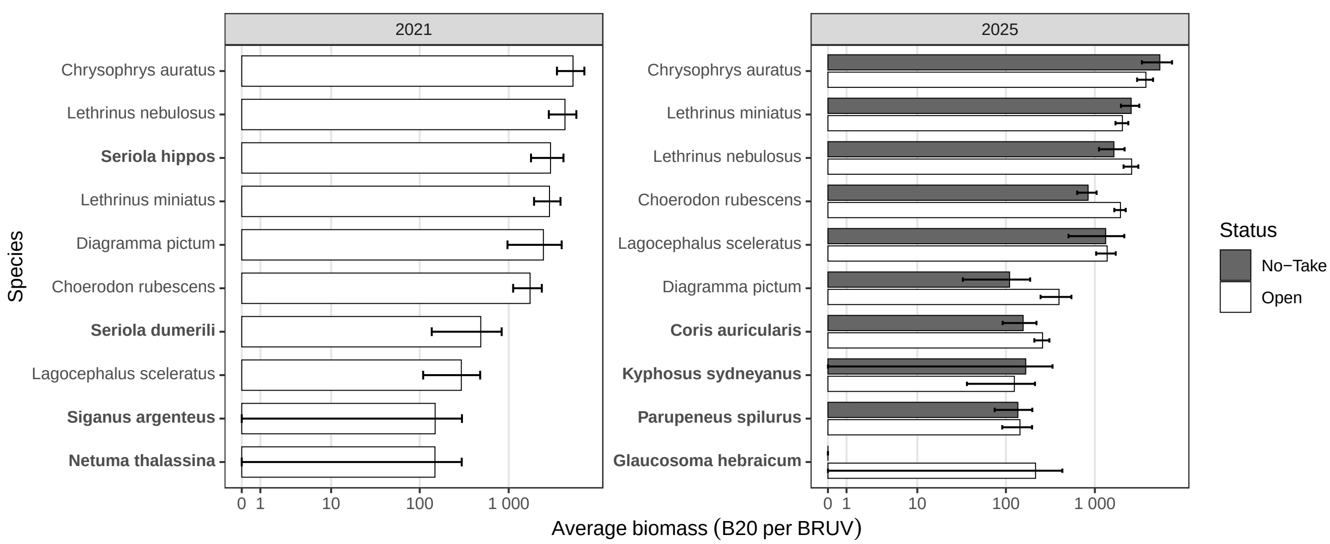

Figure 12.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species in each survey year from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). The 2021 survey is shown pooled across zones because only five deployments were in fished waters.

Large Reef Fish Index*

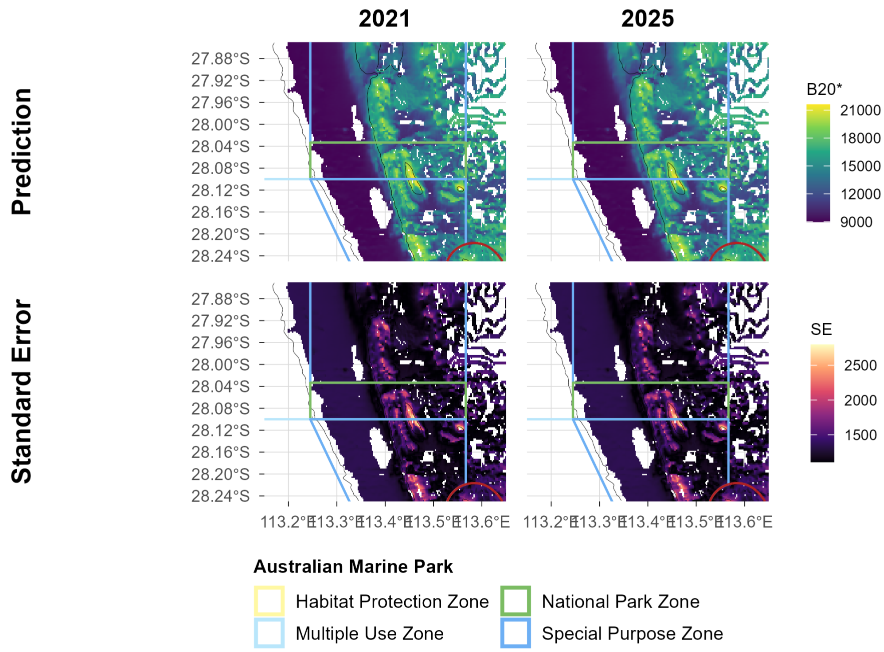

Figure 12.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

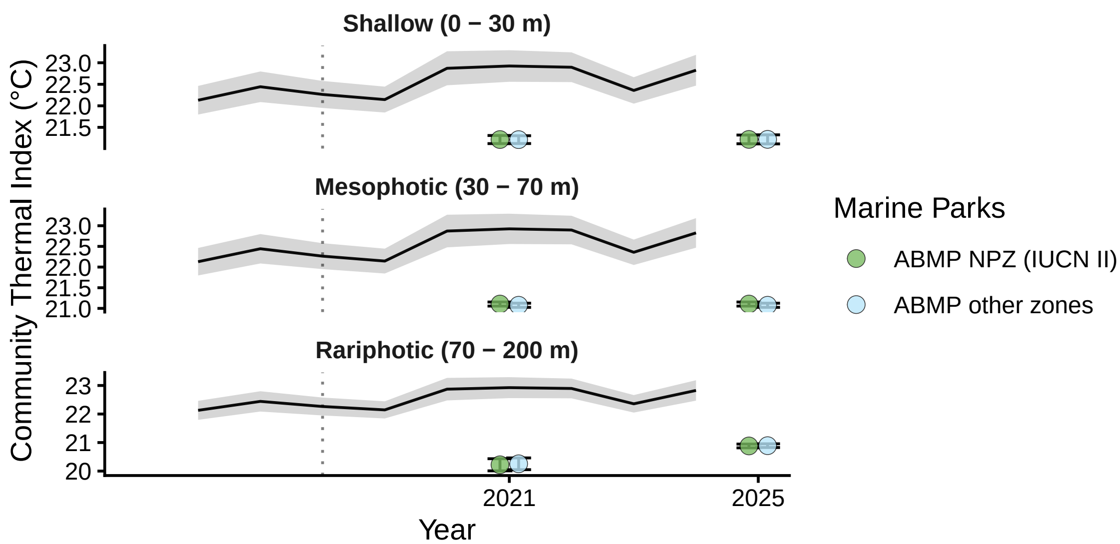

Figure 13.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with annual sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

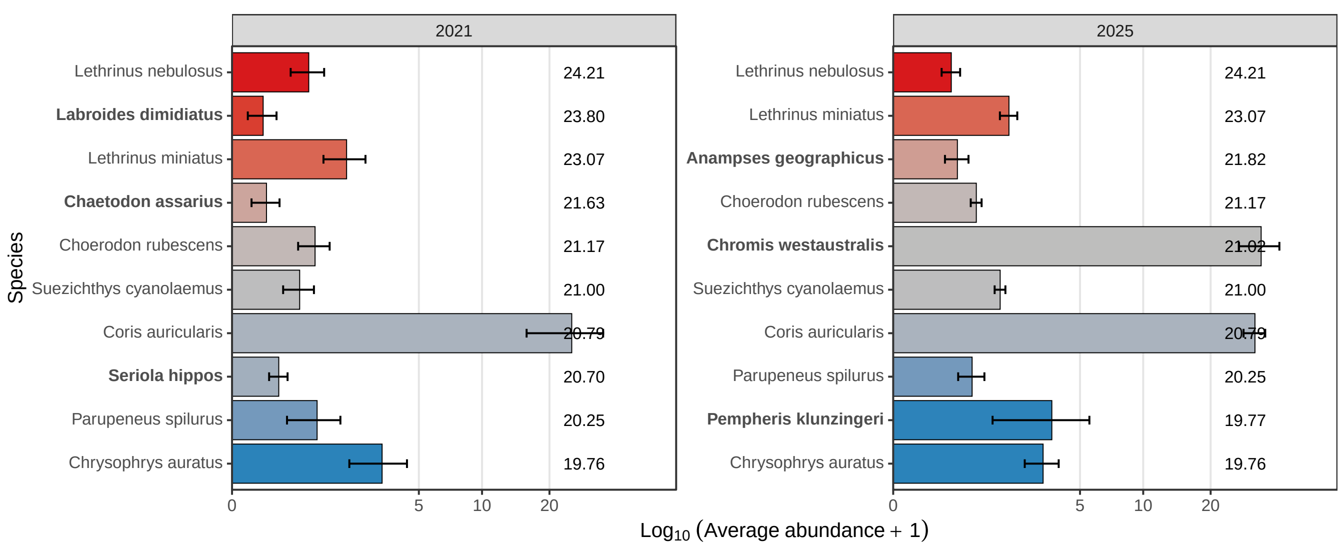

Figure 13.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of the top 10 most abundant species in each survey year from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Reef fish thermal index

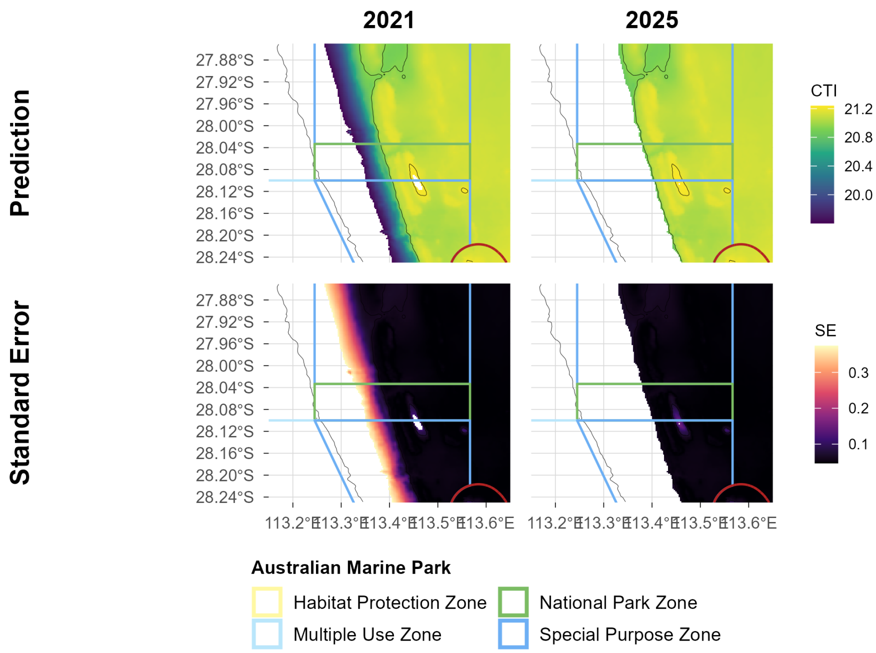

Figure 13.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing, shown separately for the 2021 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.